

Unit 6

practice K maps + Boolean

Determination of Prime Implicants

prime implicants - product term implicant which are no longer implicants if any literals are deleted from them.

Step 1. Eliminate as many literals as possible.

theorem $\rightarrow XY + XY' = X$ $\leftarrow$ this is prime implicants.
 $\uparrow$
 this is distributive theorem

Step 2. Use a prime implicant chart to select a minimum set of prime implicants.

Quine - McCluskey

this method reduces the minterm expansion of a function to obtain a minimum sum of products.

Forming Prime Implicants

$$XY + XY' = X$$

$\uparrow$ a product of literals
 $\uparrow$ single variable

Two minterms will combine if they differ in exactly one variable

Within prime implicants-

To reduce the required number of comparisons, the binary minterms are sorted into groups according to the number of 1's in each term.

$$f(a, b, c, d) = \sum m(0, 1, 2, 5, 6, 7, 8, 9, 10, 14) \quad (6-2)$$

is represented by the following list of minterms:

group 0	0 0000 $\leftarrow$ no 1's
group 1	$\left\{ \begin{array}{l} 1 \ 0001 \\ 2 \ 0010 \\ 8 \ 1000 \end{array} \right\}$ 1 1's in the group
group 2	$\left\{ \begin{array}{l} 5 \ 0101 \\ 6 \ 0110 \\ 9 \ 1001 \\ 10 \ 1010 \end{array} \right\}$ 2 1's in the group
group 3	$\left\{ \begin{array}{l} 7 \ 0111 \\ 14 \ 1110 \end{array} \right\}$ 3 1's in the group

When 2 terms combine the corresponding decimal number differ by a power of 2.

Number from group 1 - group 0 will have a power of 2.
 group 2 - group 1

Can only compare groups that are adjacent to each other. This is because 2 terms can be combined if they differ in exactly 1 variable.

example:

$$\begin{array}{r} 0000 \\ + 0001 \\ \hline 0001 \end{array} \rightarrow a'b'c'd' + a'b'c'd$$

$$\rightarrow a'b'c'(d' + d) \rightarrow a'b'c'$$

$\uparrow$
 this means that the variable has change and is changing from 0 to 1.

Note: Variables that are changing more than once can't be combined.

Example of Qm combining

$$f(a, b, c, d) = \sum m(0, 1, 2, 5, 6, 7, 8, 9, 10, 14)$$

	Column 1	Column 2
G0	0 0000	0,1 000- 0,2 00-0 0,8 -000
G1	1 0001 2 0010 8 1000	1,5 0-01 1,9 -001 2,6 0-10 2,10 -010 8,9 100- 8,10 10-0
G2	5 0101 6 0110 9 1001 10 1010	5,7 01-1 6,7 011- 8,14 -110 10,14 1-10
G3	7 0111 14 1110	

Notice for combining in group 2 that most of the numbers match up

$$7-5=2 \rightarrow 2'$$

Note: A term can be used more than once between $X + X = X$

Example:

$$\begin{array}{r} 10-10 \\ 10-11 \\ \hline 10-1- \end{array} \leftarrow \text{only variable is changing}$$

ab'd

the terms without checks are prime implicants

	Column 1	Column 2	Column 3
G0	0 0000 ✓	0,1 000- ✓ 0,2 00-0 ✓ 0,8 -000 ✓	0,1,8,9 -00- 0,2,8,10 -0-0 0,8,9 -00- 0,8,2,10 -0-0
G1	1 0001 ✓ 2 0010 ✓ 8 1000 ✓	1,5 0-01 X 1,9 -001 ✓ 2,6 0-10 ✓ 2,10 -010 ✓ 8,9 100- ✓ 8,10 10-0 ✓	
G2	5 0101 ✓ 6 0110 ✓ 9 1001 ✓ 10 1010 ✓	5,7 01-1 X 6,7 011- X 8,14 -110 ✓ 10,14 1-10 ✓	2,6,10,14 --10 2,10,6,14 --10
G3	7 0111 ✓ 14 1110 ✓		

Unit 6 problems

6.2(a)

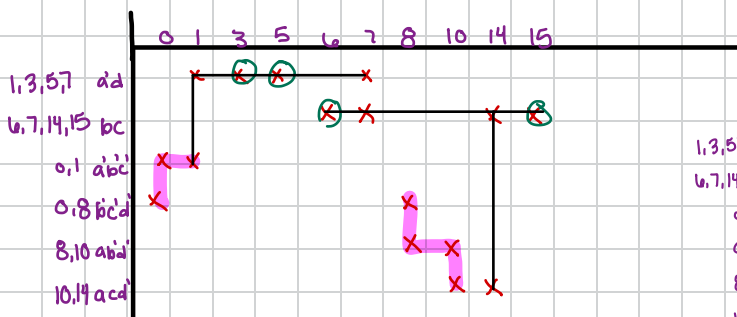
$$F(A,B,C,D) = \sum m(1,5,7,9,11,12,14,15)$$

$g1 = 0001 \downarrow$
 $g2 = 0101 \downarrow$
 $g3 = 0111 \downarrow$
 $g4 = 1011 \downarrow$
 $g5 = 1101 \downarrow$
 $g6 = 1111 \downarrow$

1 0001 ✓	1,5 0-01 → A'C'D
5 0101 ✓	1,9 -001 → B'C'D
9 1001 ✓	5,7 01-1 → A'BD
12 1100 ✓	9,11 10-1 → AB'D
7 0111 ✓	14,12 11-1 → ABD
11 1011 ✓	13,7 -111 → BCD
14 1110 ✓	15,11 1-11 → ACD
15 1111 ✓	15,14 111- → ABC

Note: the ones with prime implicants must stay throughout all of the cases.

6.3 b)



case 1. $f = d'd + bc + a'b'c' + ab'd'$

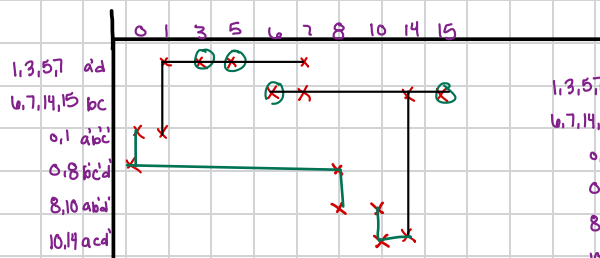
6.2 b)

$$f(a,b,c,d) = \sum m(0,1,3,5,6,7,8,10,14,15)$$

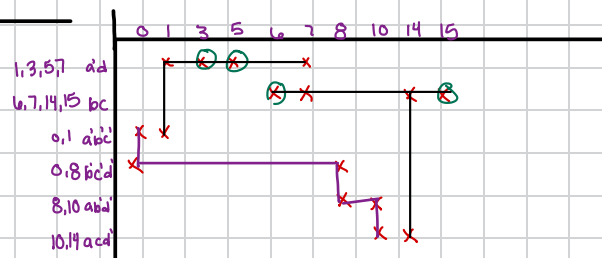
$0000 \downarrow$
 $0001 \downarrow$
 $0101 \downarrow$
 $0110 \downarrow$
 $0111 \downarrow$
 $1000 \downarrow$
 $1010 \downarrow$
 $1110 \downarrow$
 $1111 \downarrow$

0 0000 ✓	0,1 → 000- A'B'C'	3,1;7,5 → 0--1 → A'D
1 0001 ✓	0,8 → -000 B'C'D	5,1;7,5 → 0--1
8 1000 ✓	3,1 → 00-1 A'BD ✓	
3 0011 ✓	5,1 → 0-01 A'C'D ✓	7,6;14,5 → -11- BC
5 0101 ✓	10,8 → 10-0 A'B'D	11,6;13,7 → -11-
6 0110 ✓	7,3 → 0-11 A'CD ✓	
10 1010 ✓	7,5 → 01-1 A'BD ✓	
7 0111 ✓	7,6 → 011- ABC ✓	
14 1110 ✓	14,6 → -110 BCD' ✓	
15 1111 ✓	14,10 → 1-10 ACD' ✓	
	15,7 → -111 BCD ✓	
	15,14 → 111- ABC ✓	

$$A'B'C' + B'C'D' + A'B'D' + ACD' + A'D + BC$$



Case 2. $f = a'd + bc + b'c'd' + acd'$



Case 3. $f = a'd + bc + b'c'd' + ab'd'$